

Study questions for oral exam on first course in mathematical statistics, MASA02

1 Inference Theory

1. State the inference problem, based on data. Let $x_1, \dots, x_n$ be an i.i.d. sample from a random variable X with distribution function $F_\theta \in \{F_\theta : \theta \in \Theta\}$. Define the concepts of parametric, non-parametric and semi-parametric inference problems.
2. Define a statistic.
3. Which are the three general ways to make inference about a parameter? Exemplify these for the situation x an observation of X which is $\text{Bin}(25, \theta)$ -distributed.
4. Define the empirical distribution F_n based on a sample $x_1, \dots, x_n$. Viewing F_n as a random element, what is the distribution of $nF_n(x)$? Derive from this property, $\mathbb{E}(F_n(x))$ and $\text{Var}(F_n(x))$.
5. Let $x_1, \dots, x_n$ be i.i.d. observations of X with unknown distribution function $F \in \{F : F \text{ distribution function on } \mathbb{R}\}$. For $g : \mathbb{R} \rightarrow \mathbb{R}$ a function, describe $\eta := \mathbb{E}(g(X))$ as a functional of F . Explain for which g this definition make sense. Define the plug-in estimator of η . Apply the definition on the estimation of $\eta = \mathbb{E}(X^3)$, and on $\mathbb{E}(X \cdot 1\{X \in [0, 1]\})$.
6. What is an unbiased estimator? Define the mean square error (m.s.e.) of an estimator. Prove the partition of the m.s.e. into it's bias and variance part.
7. Define consistency of an estimator, in quadratic mean, almost surely and in probability. Show that consistency in quadratic mean implies consistency in probability.
8. Show that the plug-in estimator of a functional of the form $\theta = \int g(x)dF(x)$ is consistent in quadratic mean.
9. Let $x_1, \dots, x_n$ be a sample from a distribution F_θ , with $\theta \in \Theta \subset \mathbb{R}^s$, with $s < \infty$. Describe the maximum likelihood method for estimating a parameter θ . Apply the method for estimating $\theta \in \Theta = [0, \infty)$ when one has one observation x of X which is $Un[0, \theta]$ -distributed.
10. Let $x_1, \dots, x_n$ be a sample from $N(m, \sigma^2)$, with m and σ^2 unknown. Derive the maximum likelihood estimators of μ and σ^2 . Let x be a sample from $\text{Bin}(n, p)$ with p unknown. Derive the maximum likelihood estimator of p .
11. Let $x_1, \dots, x_n$ be a sample from a distribution F_θ , with $\theta \in \Theta \subset \mathbb{R}$ unknown. Describe the least-squares method (ls) for estimating a parameter θ .

12. Derive the ls estimators for the parameters m and p in question 10.
13. Describe the idea behind a confidence interval for an unknown one-dimensional parameter θ in a continuous distribution, with a certain confidence grade. Describe the method with pivot function or pivot variable for constructing a confidence interval.
14. Describe the idea underlying a statistical test. Define the concepts null- and alternative hypothesis, test statistics, critical region and significance level. Describe the method with pivot function or pivot variable for constructing tests for a one-dimensional parameter in a continuous distribution. Define the power function for a given test.
15. Describe the p-value method for testing.
16. Suppose that θ is the unknown parameter in a *continuous* distribution. Can you use both the p-value method and the method with critical regions for testing? Suppose that θ is the unknown parameter in a *discrete* distribution. Can you use both the p-value method and the method with critical regions for testing?
17. Let $x_1, \dots, x_n$ be a sample from $N(m, \sigma^2)$, with m unknown, σ^2 known. Derive a 95% two-sided confidence interval m . Derive a test on significance level 5% for the hypotheses $H_0 : m \geq 1$, $H_1 : m < 1$.
18. (ctd.) Derive the power function of the test in the previous question.
19. Suppose that $x_1, \dots, x_n$ is sample from $N(m, \sigma^2)$, with both m and σ^2 unknown. Derive a 99% one-sided lower bounded confidence interval for m . Derive a test on significance level 1% for the hypotheses $H_0 : m = 2$, $H_1 : m \neq 2$.
20. Suppose that $\theta = (\theta_1, \dots, \theta_k)$ is a k -dimensional parameter and suppose that somehow you have derived, or you have been given, k confidence intervals $I_1, \dots, I_k$, with each I_j a confidence interval for θ_j with the same confidence grade $1 - \alpha$. Is $I = I_1 \times \dots \times I_k$ a (simultaneous) confidence set for θ , with confidence grade $1 - \alpha$? Why not? Give a lower bound on the confidence grade for the simultaneous confidence set I .
21. Describe the problem with repeated tests when testing for a parameter $\theta = (\theta_1, \dots, \theta_k)$. Describe the Holm-Bonferroni correction method for multiple testing.
22. Describe the confidence method for doing tests. Describe how to do test-based confidence intervals.
23. If X is a continuous random variable with density, why can you not use the plug-in approach to estimating $\eta = T(F) = f(x_0)$, for a fixed x_0 ? Define the kernel estimator of $\eta = f(x_0)$. Why is the kernel estimator preferable to the histogram?
24. How do you construct non-parametric one-sample tests with the use of the empirical distribution function? How do you construct confidence bands for F ? How do you construct a k -sample test?
25. Suppose that θ is the linear functional in problem 8, and let $\hat{\theta}_n$ be the plug-in estimator. Suppose that $\text{Var}(g(X)) = \sigma^2 < \infty$ is known. What is the limit distribution of $\hat{\theta}_n$? Derive an approximate 95% confidence interval for $\mathbb{E}(g(X))$. What happens when σ^2 is not known?

26. State the continuous mapping theorem and Slutsky's theorem. Give one situation in which are they used?
27. Suppose that $(x_{1i}, \dots, x_{pi}, y_i), i = 1, \dots, n$ are observations in the regression problem

$$y_i = \theta_0 + \theta_1(x_{1i} - \bar{x}_{1.}) + \dots + \theta_p(x_{pi} - \bar{x}_{p.}) + \epsilon_i,$$

- with $\epsilon_i \in N(0, \sigma^2)$ i.i.d. random variable. Write the regression problem on matrix form. Derive the least squares estimator $\hat{\theta}$ of the vector $\theta = (\theta_0, \dots, \theta_p)$. Show that $\hat{\theta}$ is unbiased. Derive the covariance matrix of $\hat{\theta}$. What is the distribution of $\hat{\theta}$?
28. (ctd.) Explain how you can use this knowledge to make tests and confidence intervals for the θ -parameters in the previous question, when σ^2 is known. If $a_1, \dots, a_p$ are known and $\eta = a_1\theta_1 + \dots + a_p\theta_p$ a parameter of interest, how do you make tests and construct confidence intervals for η ? What property of the multivariate Gaussian distribution do you use for this?
29. (ctd.) Derive the maximum likelihood estimator for θ in the previous question. Derive the maximum likelihood estimator of σ^2 . Explain how you can make tests and confidence intervals for θ when σ^2 is unknown.